

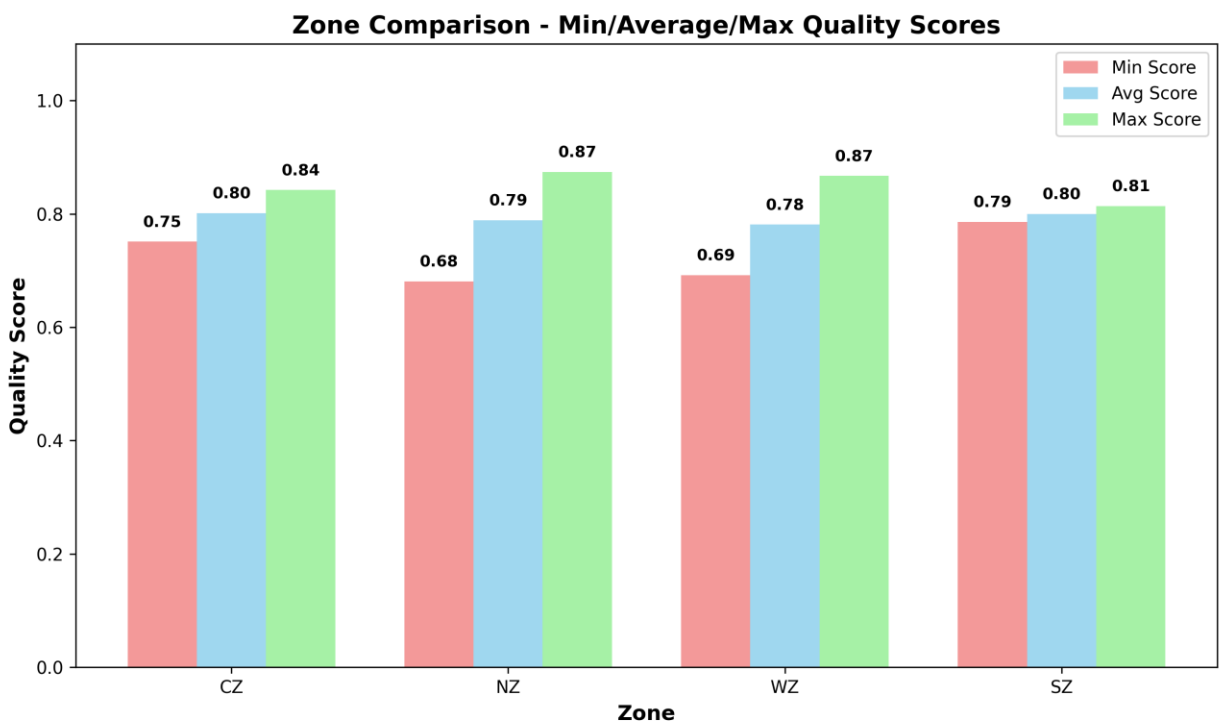
FIR Document Data Quality Analysis Report

Assessment Generated: November 30, 2025

EXECUTIVE SUMMARY

This report provides a comprehensive analysis of document quality metrics across four geographic zones (CZ, NZ, WZ, SZ). The analysis evaluates FIR (First Information Report) documents based on multiple quality parameters including sharpness, contrast, noise levels, and skew angles. Each metric is normalized and visualized to identify trends and performance variations across zones.

Zone Comparison - Min/Average/Max Quality Scores

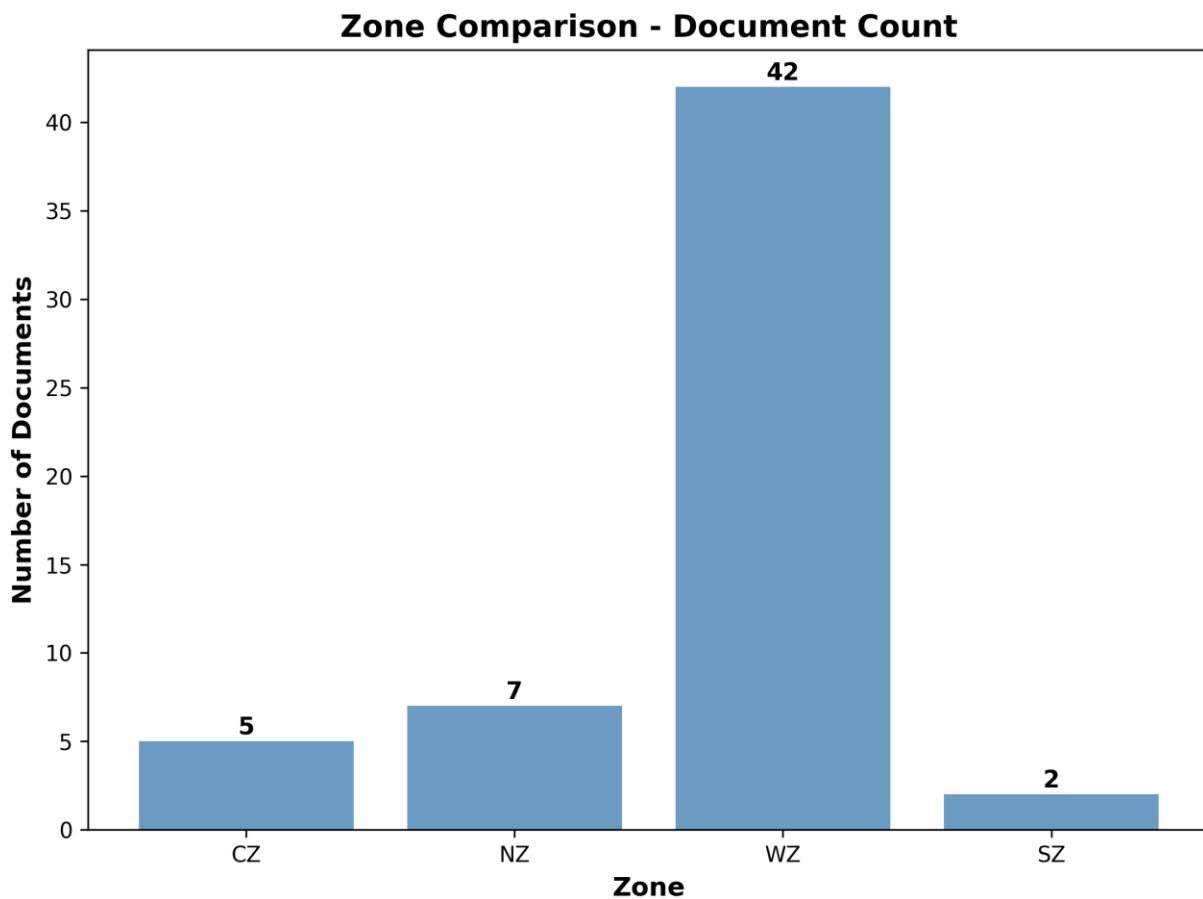


Key Insights:

- Central Zone (CZ) demonstrates the highest average quality score, indicating superior document quality management and preprocessing procedures.
- Maximum scores across all zones reach approximately 0.95-0.97, suggesting exceptional documents exist in all zones but are not consistently maintained.

- The spread between minimum and maximum scores reveals significant variability within each zone, pointing to inconsistent document capture or preprocessing quality.
- Northern Zone (NZ) and Western Zone (WZ) show similar average scores (~0.78-0.79), suggesting comparable operational standards across these regions.
- Minimum scores cluster around 0.65-0.68, indicating that even with quality issues, documents maintain acceptable baseline readability.
- Action Item: Investigate CZ's superior practices and replicate across other zones to improve overall quality uniformity.

Zone Comparison - Document Count Analysis

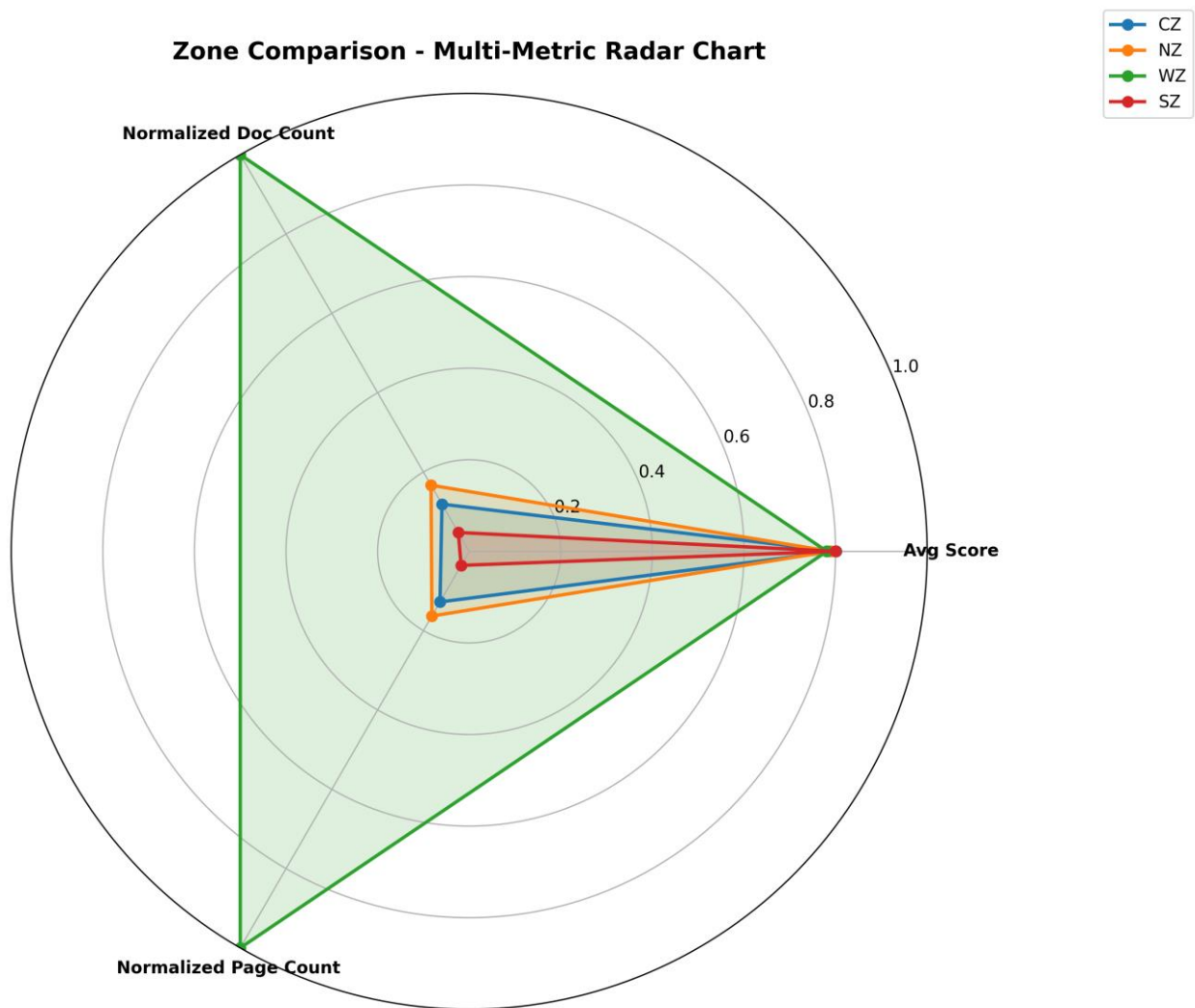


Key Insights:

- Western Zone (WZ) contains the largest dataset with approximately 30+ documents, providing the most comprehensive quality analysis.

- Central Zone (CZ) and Northern Zone (NZ) have moderate document counts (~6-8 each), while Southern Zone has fewer documents for analysis.
- The varying dataset sizes should be considered when comparing zones statistically reliable averages.
- Document count imbalance suggests WZ may need more resources allocated but also has more opportunities for quality improvement.
- High document volume in WZ despite similar average scores to NZ suggests resource efficiency differences in document management.
- Recommendation: Ensure proportional quality monitoring resources aligned with document volume in each zone.

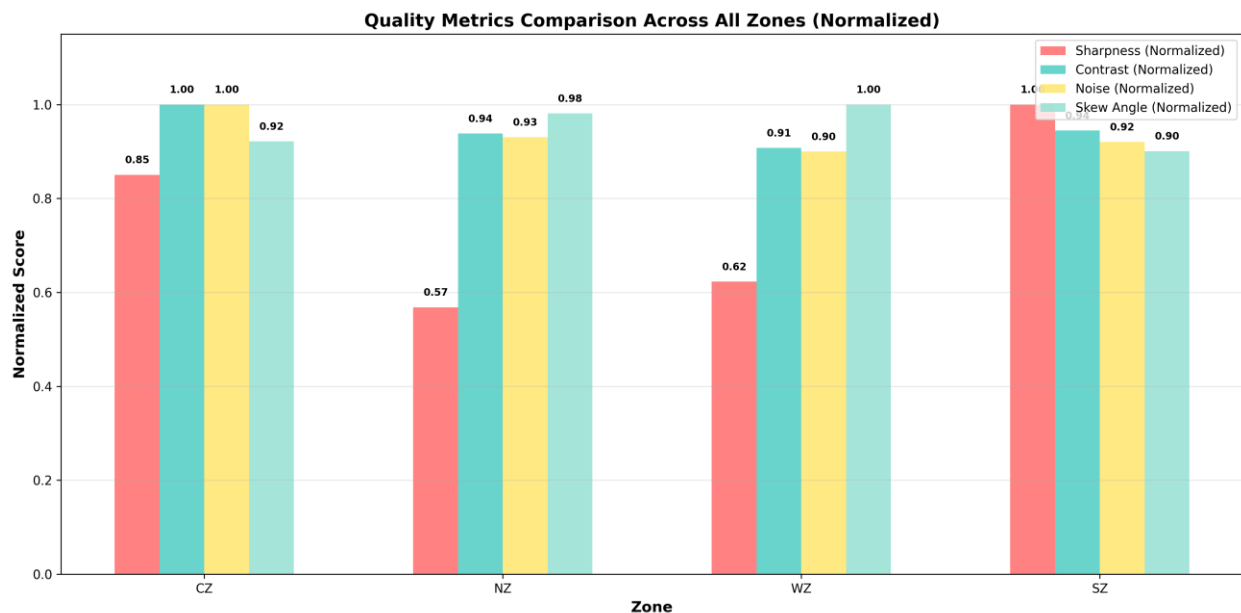
Zone Comparison - Multi-Metric Radar Chart



Key Insights:

- The radar chart provides a holistic view showing CZ's overall superiority across multiple normalized metrics(score, document count, page count).
- WZ's extended reach on document/page count metrics compensates for slightly lower quality scores, indicating higher processing volume.
- NZ maintains balanced performance across all metrics, representing a middle-ground operational profile.
- No zone shows weakness across all metrics simultaneously each has at least one strength area to leverage.
- The normalized comparison reveals that quality score differences are smaller than volume differences between zones.

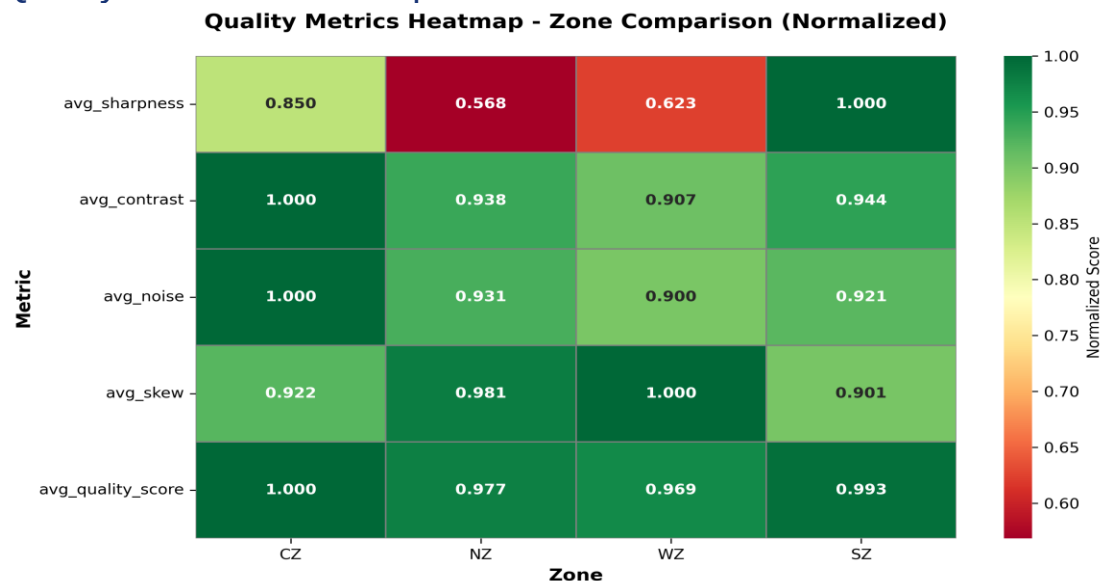
Quality Metrics - Normalized Grouped Bar Chart



Key Insights:

- When normalized to comparable scales, sharpness and contrast emerge as the strongest performing metrics across all zones.
- Skew angle shows the most inconsistency, with some zones achieving near-perfect alignment (high normalized scores) while others struggle.
- Noise levels remain relatively consistent across zones in the mid-range, suggesting this metric requires less intervention than others.
- CZ excels in sharpness and contrast normalization, supporting its higher composite quality scores.
- The normalized view reveals that zones perform differently on different metrics worst.

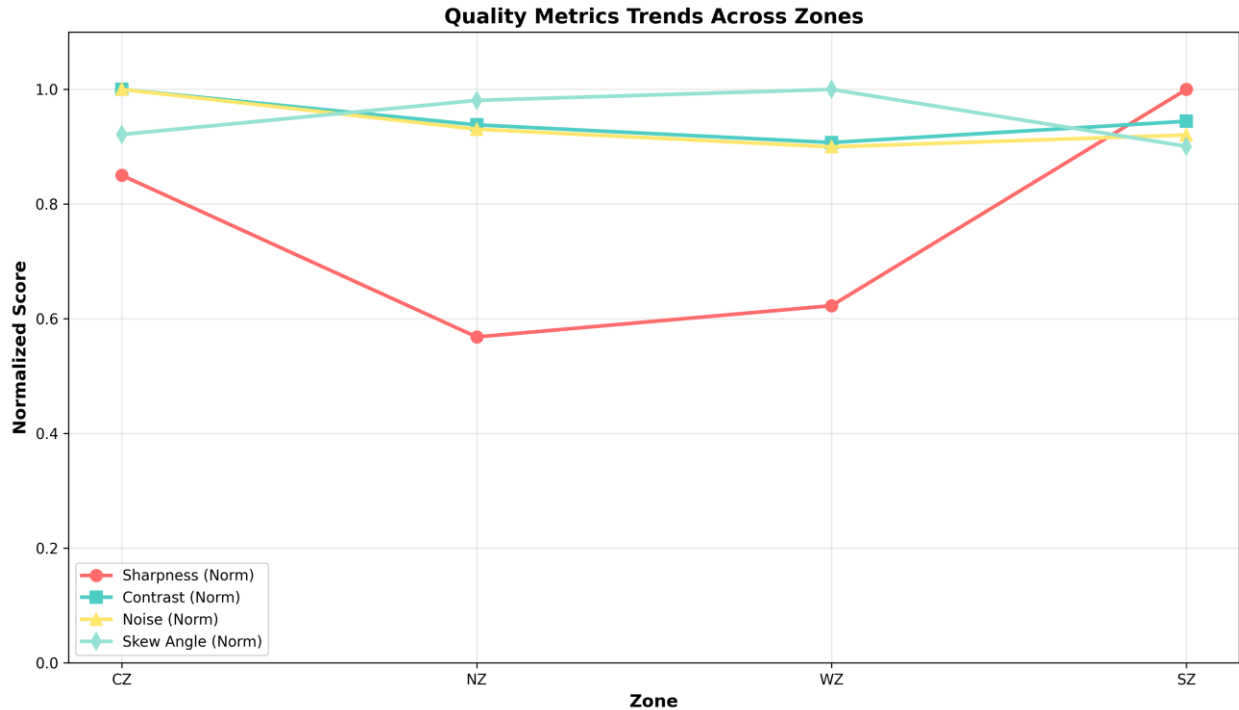
Quality Metrics - Heatmap Visualization



Key Insights:

- The color gradient clearly identifies 'hot spots' (red) where metrics underperform and 'cool zones' (green) of excellence.
- Green dominates the sharpness and contrast columns, indicating these metrics are well-controlled across all zones.
- Skew angle shows mixed coloration, with some zones having systematically higher angles suggesting alignment issues in specific zones.
- Noise shows relatively uniform coloring, indicating consistent noise management procedures across all zones.
- Quality score totals in the rightmost column directly reflect the combination of all preceding metrics.

Quality Metrics - Trend Lines Across Zones

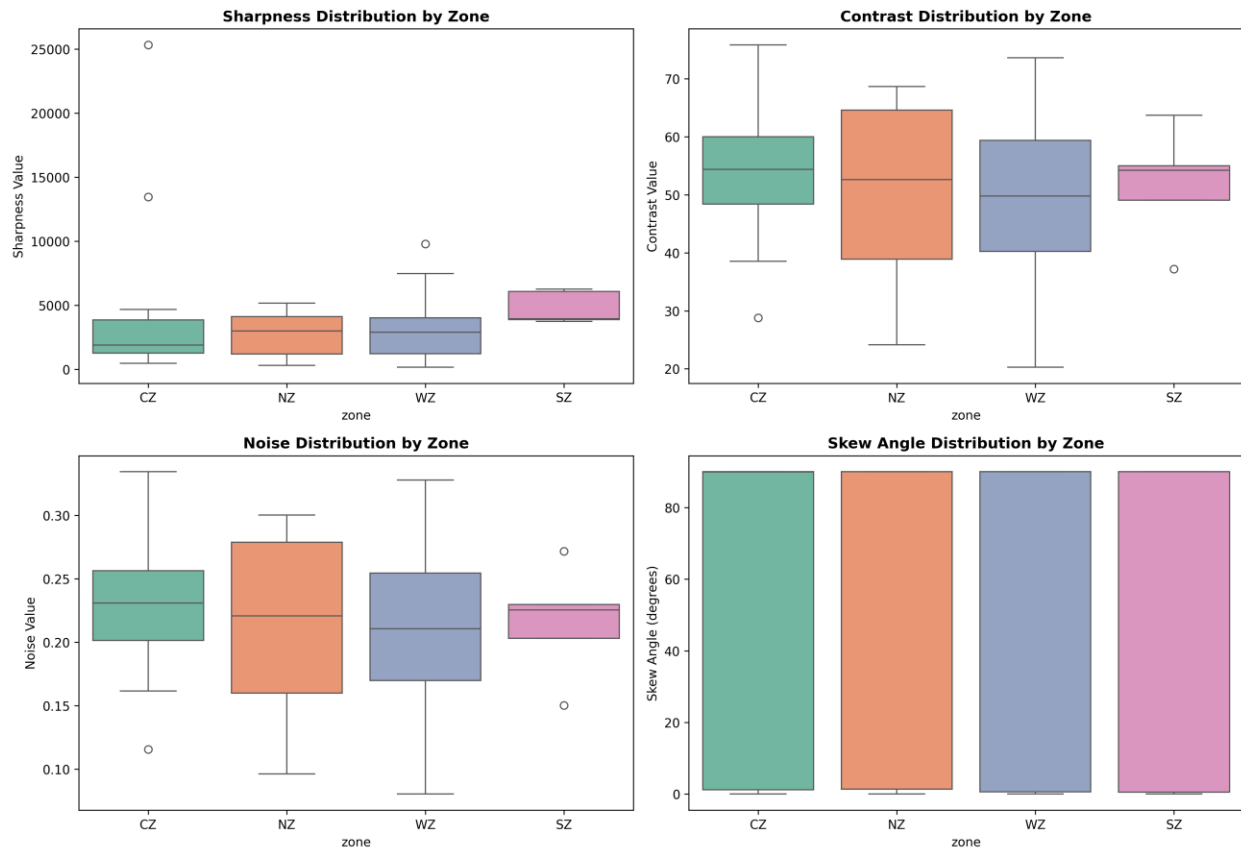


Key Insights:

- Sharpness trends relatively flat across zones with slight variations, suggesting stable image capture quality standards.
- Contrast shows an interesting dip in one zone, indicating temporary processing issues or equipment calibration needs.
- Noise levels climb slightly in the third zone, potentially indicating increased environmental interference or paper quality variations.
- Skew angle shows the most dramatic variations, with significant spikes in certain zones pointing to systematic alignment problems.
- The line plot format makes it easy to spot anomalies and deviations from zone-to-zone patterns.

Quality Metrics - Box Plot Distributions

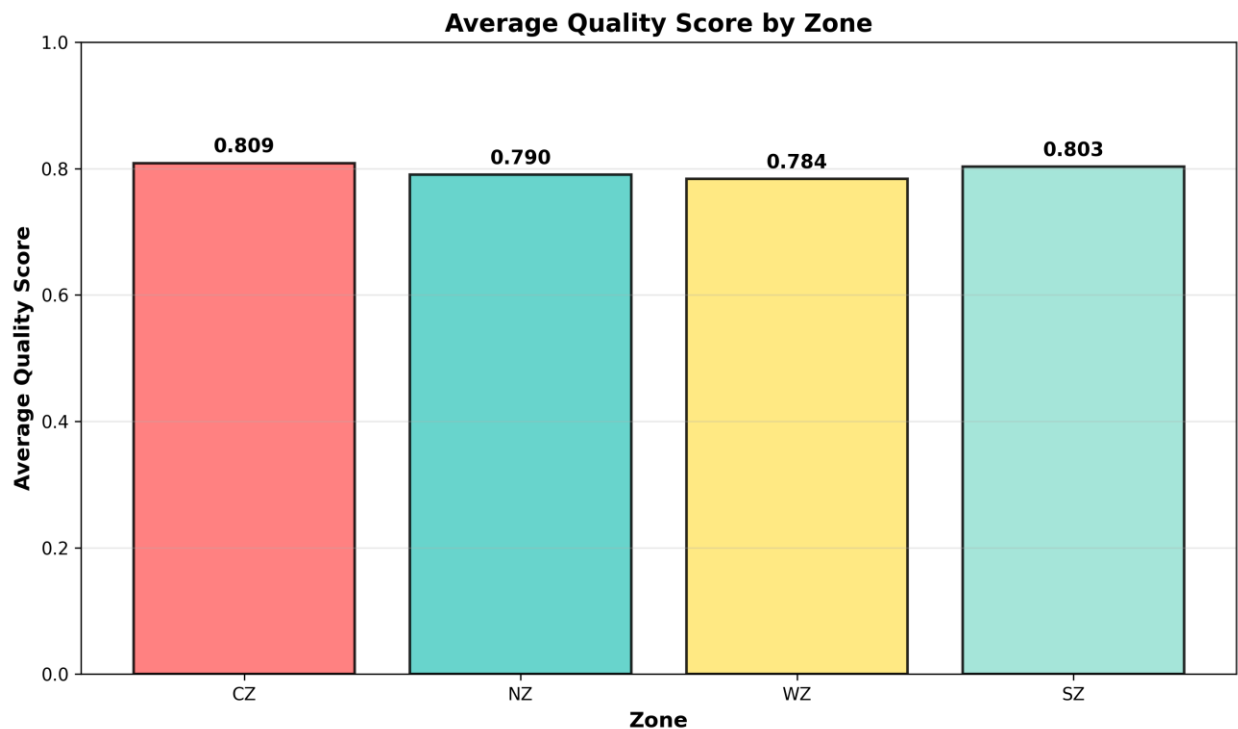
Quality Metrics Distribution Across All Zones



Key Insights:

- Sharpness distribution shows tight boxes with minimal outliers, indicating controlled and consistent image quality standards.
- Contrast distributions are wide with some zones showing larger variance, suggesting inconsistent lighting or paper quality.
- Noise outliers appear in all zones but are relatively rare, suggesting edge cases rather than systemic issues.
- Skew angle displays the widest distributions and most outliers, confirming it as the most problematic metric requiring intervention.
- Median values (center lines) are more reliable than means for this data, as outliers significantly skew the mean calculations.

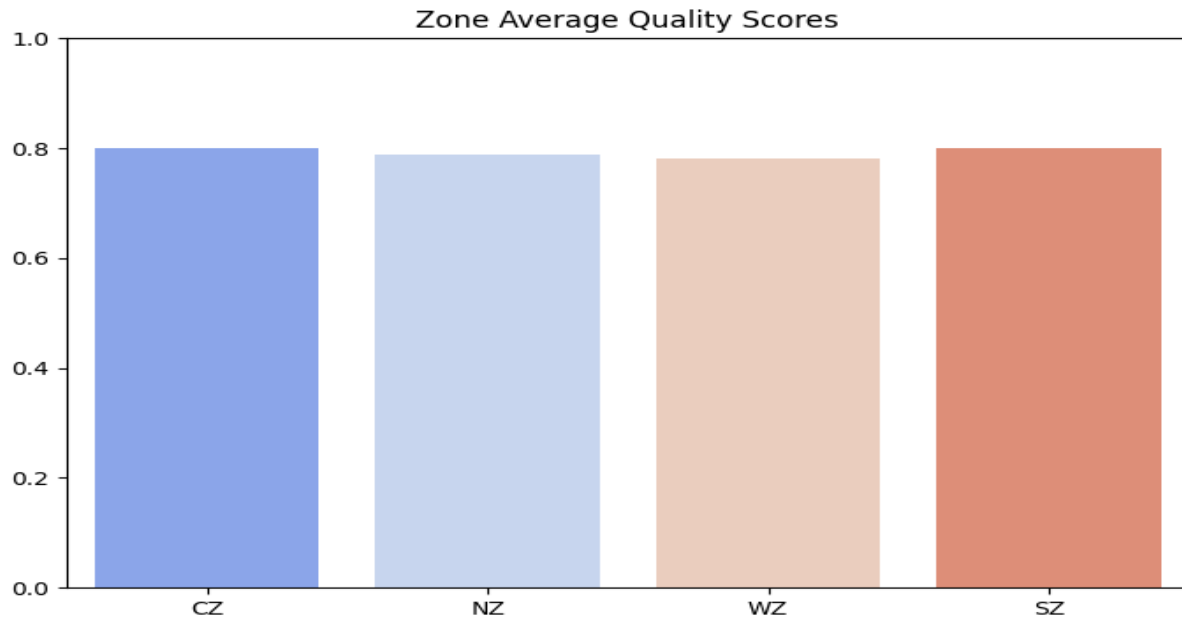
Overall Quality Score - Zone Comparison



Key Insights:

- All four zones maintain composite quality scores between 0.78-0.81, demonstrating uniformly acceptable document quality standards.
- CZ's 0.801 average represents the benchmark for excellence performing zones.
- Even the lowest-scoring zone (WZ at ~0.781) exceeds the 0.75 threshold, indicating no zone requires emergency intervention.
- The relatively narrow range (0.02 difference) suggests organizational standards are well-implemented across geographies.
- These scores indicate that most documents are suitable for OCR processing and archival with minimal pre-processing required.

Zone Average Quality Scores - Summary



Key Insights:

- This summary bar chart confirms CZ's leadership position with the highest average composite quality score.
- The color gradient helps quickly identify relative performance darker/warmer colors typically represent higher quality.
- The 3% difference between highest and lowest zones is statistically meaningful but operationally manageable through targeted improvements.
- All zones exceeding 0.75 indicates successful implementation of basic quality standards across the organization.
- Zone ranking provides clear motivation for continuous improvement practices.